

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Built scalable cloud backend services and APIs for a multi-tenant NHS eTriage system (Node.js, MongoDB) serving **8+ UK clinical clients** and **10,000+ monthly submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Designed and coded a full-stack React web portal used daily by the AI team to structure and label ML training data, cutting manual annotation by **100+ hours/month** and lifting throughput by **60%**.
- Owned backend reliability and quality: added integration tests, monitored production logs and debugged incidents, hardened auth and rate limiting with **AWS KMS** encryption, and remediated **20+ VAPT findings**.
- Served as primary technical POC, ran knowledge transfer onboarding a second developer in **~6 weeks**, and piloted a COO-sponsored GitHub Copilot experiment for AI-assisted development.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful web services** (Node.js, Fastify, PostgreSQL) with validation, clear client-server contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core repositories from JavaScript to TypeScript and shipped a production LLM feature (**4-bit quantized Llama-7B via llama.cpp**) with async queued generation and content-safety filtering.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform with git-webhook auto-deploy (CI/CD), cron automation, and Stripe payments; live **3+ years**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a cloud-deployed Python/FastAPI backend service that uses ML/AI to turn open-ended queries into structured, verifiable entity tables via typed retrieval planning, reranking, and deterministic extraction; hardened with **150+ tests**.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built a full-stack edge web app with a chat UI over serverless cloud services (Cloudflare Workers AI, Durable Objects, D1), streaming AI responses with persistent per-session state and a typed client-server API.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Optimized a large-scale data-processing pipeline over **~1.4B rows**, cutting end-to-end runtime **~25%** and the tail-latency ratio **2.14x to 1.74x** via broadcast joins, AQE, skew handling, and partitioning.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: JavaScript, TypeScript, Python, C++, Java, SQL, Bash, PostgreSQL, MongoDB, SQLite

Frameworks / Tools: Node.js, Express, React, FastAPI, Docker, Git, Cloudflare Workers, REST APIs

DevOps / Testing: Linux, CI/CD, GitHub Actions, unit & integration testing, monitoring, shell scripting, automation

ML & AI: PyTorch, Hugging Face Transformers, LLM integration, RAG, analytics pipelines, evaluation

Cloud / Platforms: AWS, GCP, Azure, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.